

MATH 317 Project 1: Autonomous Detection of Color Vision Deficiency

Maxwell Hausler
Principal author
mhausler@vassar.edu

Jack Regovich
Collaborator
jregovich@vassar.edu

Abstract—Our problem is to detect the colors which our sample patients are most likely to confuse by modifying our stimulus color to minimize the steady-state visually evoked potential or SSVEP response at 5 Hertz and processing our data from the amplitudes of the electroencephalograph or EEG. To do so, I approximated the gradient using the amplitudes given from the frequency spectrum of the EEG data and utilized them in a forward finite difference approximation of the first order. With our gradient approximation, I applied the gradient descent algorithm, used a line search to improve its convergence and applied penalty functions to keep our problem within its constraints, those being that the entries representing the RGB values of our stimulus color must remain in the interval $[0,1]$ and that our algorithm doesn't converge to a solution where the stimulus color is approximately equal to the reference color in RGB. I reached the conclusion that Patient 1 confuses green for yellow and likely has deuteranomaly and that Patient 2 confuses green for blue and likely has tritanopia or tritanomaly.

I. INTRODUCTION

A. Problem Description

Subjects are shown two distinct colors, one that is a reference color and one that is a flickering stimulus color. Our brains will mimic the frequencies of the colors shown to them through electrical activity and this activity should spike at the flickering frequency to mark that the viewer sees it as distinct to the reference color. We have the EEG data from their observation of these colors and this has undergone a harmonic analysis or Fourier transform to produce a frequency spectrum and from this, we are meant to identify the amplitudes and minimize them in order to determine the colors which the patients are most likely to confuse and thus, the colors they will not see as distinct.

To phrase this as an optimization problem, we are trying to minimize a function which represents the amplitudes in a set of data and it has the first-order necessary condition of differentiability as well as the constraints that the entries of the stimulus color vector must remain in the interval $[0,1]$ and that our algorithm cannot converge to a solution where the stimulus color is approximately equal to the reference color. To optimize it, we have to take the gradient of this function and apply the gradient descent optimization algorithm to minimize the amplitudes such that $\|x_{i+1} - x_i\| < \epsilon$ and $|f(x_{i+1}) - f(x_i)| < \delta$.

II. DESCRIPTION OF NUMERICAL METHODS USED

A. Gradient Descent

Gradient descent is an optimization algorithm which executes for $\mathbb{R}^n \rightarrow \mathbb{R}$ and thus, it has a vector as its input and a scalar as its output. Gradient descent takes an initial guess x_0 , iterates and has the general rule $x_{i+1} = x_i - \alpha \nabla f(x_i)^T$ where α is our step size. This algorithm works according to the idea that our gradient represents the direction of steepest descent and so if we step in that direction, we will get closer to a local minimum and thus, a point we can optimize our function at.

B. Finite Difference Approximation

I used a forward finite difference approximation of the first order which has the form $\frac{f(x+h) - f(x)}{h}$. This comes from our limit definition of a derivative as h , our step size, approaches zero and thus, our finite difference approximation approximates the derivative and improves in accuracy as h gets smaller. This method is optimal since we do not have a general function for our data but rather just a finite set of data of the amplitudes and so we cannot apply a general derivative rule but we can approximate it.

C. Penalty Functions

Penalty functions provide constraints for our problem. In this case, they provide the constraints that the entries of the stimulus color vector must remain in the interval $[0,1]$ and that our algorithm cannot converge to a solution where the stimulus color is approximately equal to the reference color. The goal of our penalty functions is to eliminate the output or redirect it into the admissible region if and only if the constraint is violated. In our case, I implemented the penalty functions by having the minimum functions square our output so that it remains positive and equal to zero and by having our maximum functions subtract one from their output so that it will be zero. As a result, when our constraints are violated, we will have penalty functions equal to zero, thus eliminating our inadmissible output.

D. Line Search

Line search is an iterative technique I used to improve the convergence of our gradient descent algorithm where if our step size is greater than or equal to our tolerance, we

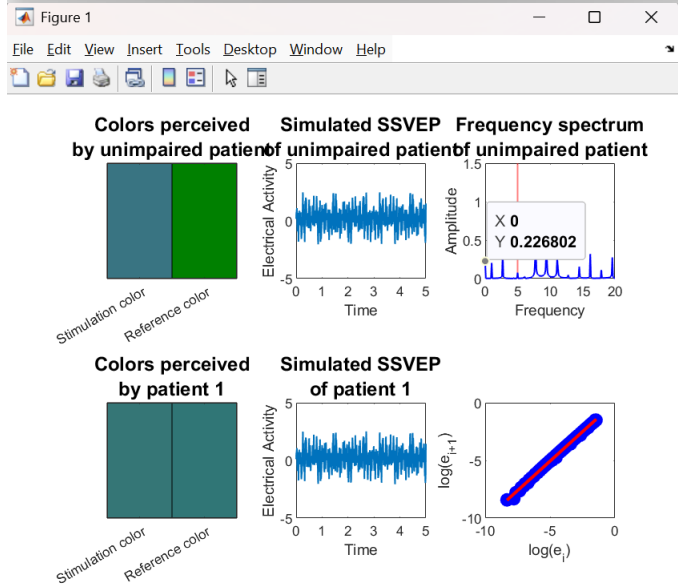
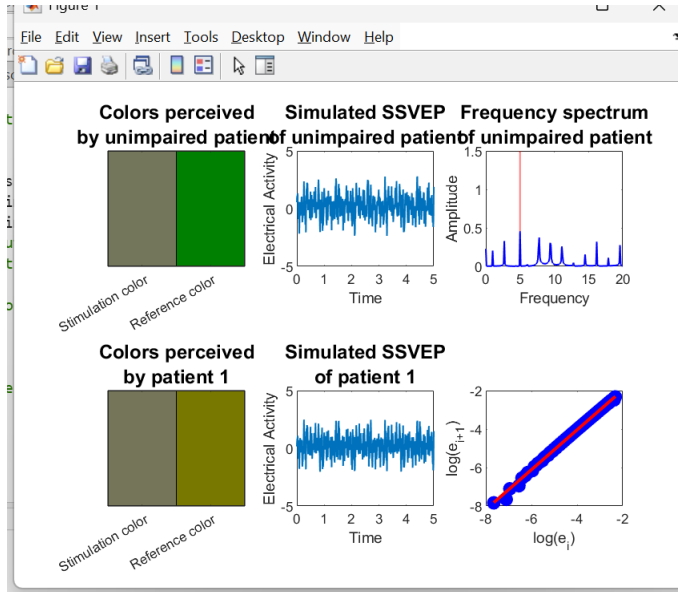
will ensure our function decreases and select the direction of steepest descent (our gradient) to continue iterating in.

E. Acceleration

I did not use acceleration for my algorithm as it was optional.

III. CONVERGENCE ANALYSIS

The order of convergence of my code is linear which is as expected since we are using a first order finite difference approximation and the gradient descent algorithm. This can be seen in the below convergence plots for Patients 1 and 2.

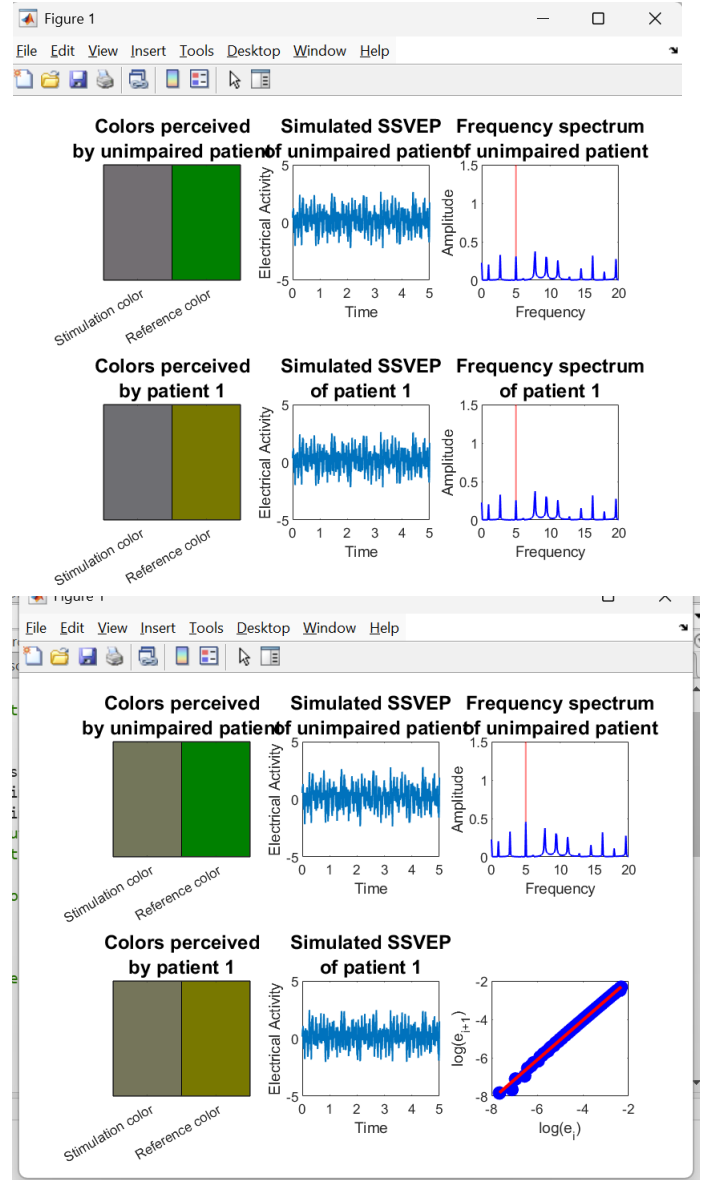


We can see from the plots that our convergence is approximately linear however there is some noise in that our convergence plot levels off around -10 which can affect our convergence however this is due to our tolerance and in

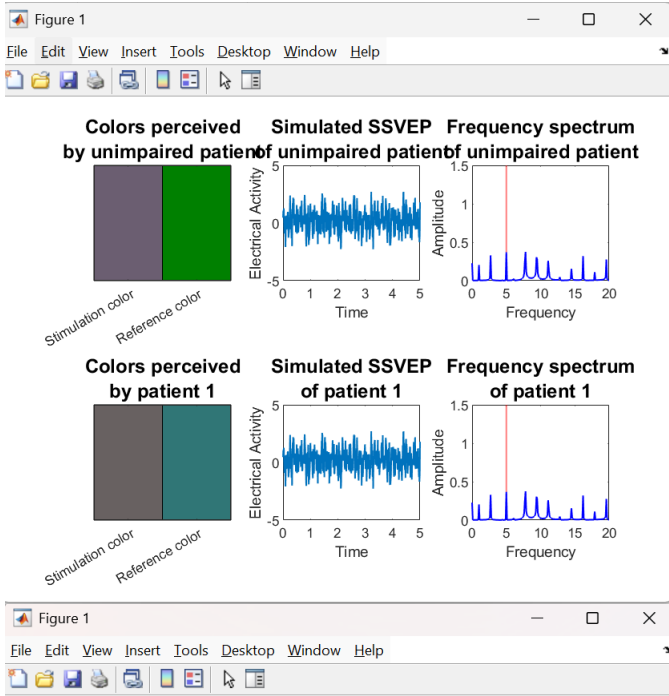
general, we see a linear convergence as predicted from both files.

IV. RESULTS

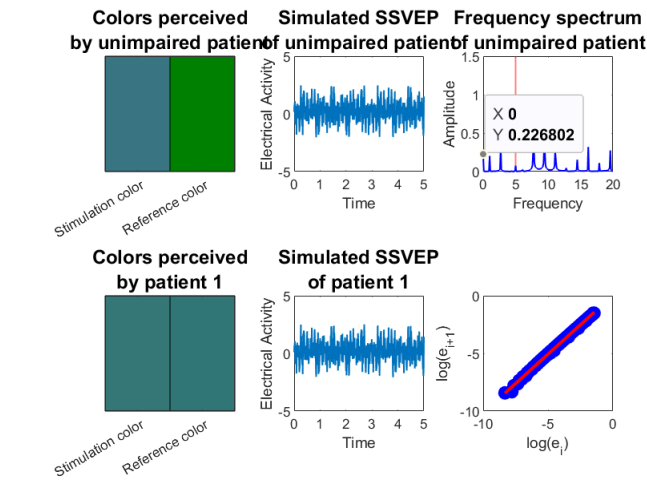
Below are the images showing the visualization simulation at the initial color and at the optimized color for each patient.



These are the images for Patient 1, they show that the patient sees green as yellow and thus, likely has deuteranomaly as a result of malfunctioning cones for processing color in their eyes.



tolerance to accommodate the noise that it generates around -10. We could also improve this approach by changing our reference color vector to be variable so as to find pairs of colors which the patients cannot distinguish rather than solely changing our stimulus color vector.



These images for Patient 2 show that the patient sees green as blue and thus, has tritanopia or tritanomaly.

V. CONCLUSION AND FUTURE DIRECTIONS

I found that Patient 1 sees green as yellow and thus, has deuteranomaly and that Patient 2 sees green as blue and thus, has tritanopia or tritanomaly. Our results come from numerous iterations of our algorithm and though they support our predictions for convergence, we could improve our approach to this optimization problem of sorts by extending our definition of the finite difference approximation to higher orders for acceleration of our algorithm for gradient descent and for smaller error. We could also extend our methods by implementing the finite difference approximation for a general case instead of our specific function and order. Additionally, we could introduce more penalty functions to reduce the number of iterations our algorithm has to execute or other accelerative techniques. It would also be useful to adjust our